Unsupervised Domain Adaptation for Cross Domain Classification of Mangrove Species

An Internship Report

submitted in partial fulfillment of the requirements of the
Summer Internship Program

by

Sikharam Satya Sri Harsha Phani

IS00988



Department of Earth and Space Science

Indian Institute of Space Science and Technology

Thiruvananthapuram, India

July 2026

Certificate

This is to certify that the internship report titled “Unsupervised Domain Adaptation for Cross-Domain Classification of Mangrove Species”, submitted by Sikharam Satya Sri Harsha Phani (IS00988), is a bona fide record of the work carried out by the intern under my supervision at the Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, during the period of the internship. The work presented in this report has not been submitted, either in whole or in part, to any other institute or organisation for the award of any degree, diploma, or certificate.

Dr. L Gnanappazham

Faculty Supervisor

Place: Thiruvananthapuram

Date: July 2026

Declaration

I hereby declare that this internship report titled “Unsupervised Domain Adaptation for Cross-Domain Classification of Mangrove Species”, submitted in partial fulfillment of the requirements of the internship program at the Indian Institute of Space Science and Technology (IIST), is a bona fide record of the original work carried out by me under the supervision of Dr. L Gnanappazham. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship, or any other academic qualification in this or any other institution. I further declare that, in keeping with ethical academic and research practices, due acknowledgment has been made wherever the work, ideas, or findings of others have been cited, and that the datasets used in this study were provided as part of the internship project.

Place: Thiruvananthapuram

Date: July 2026

Sikharam Satya Sri Harsha Phani

IS00988

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Sikharam Satya Sri Harsha Phani

Abstract

Mangrove ecosystems are highly heterogeneous, and classifiers trained on one region typically generalise poorly when applied to another due to differences in species composition, canopy structure, tidal regime, and sensor or acquisition conditions — a phenomenon broadly known as domain shift. This report presents a study on unsupervised domain adaptation (UDA) for cross-domain classification of mangrove species, carried out using pre-processed 64-band feature embeddings covering three mangrove regions of India: the Krishna delta (source domain, with labelled ground truth), and the Malad creek and Pichavaram mangrove forests (unlabelled target domains). In the first phase of the study, four shallow domain alignment strategies — Subspace Alignment (SA), Correlation Alignment (CORAL), and their two sequential hybrids (CORAL→SA and SA→CORAL) — were benchmarked on a temporal domain-shift task (Krishna 2019 versus Krishna 2024) using covariance gap and subspace distance as quantitative alignment metrics. CORAL was found to deliver the strongest joint reduction in both metrics and was selected as the domain alignment method for the subsequent cross-region classification task. In the second phase, a one-vs-rest Random Forest classifier trained on Krishna ground-truth samples was applied, after CORAL-based feature alignment, to classify mangrove species in the Malad and Pichavaram study areas. The Krishna→Malad transfer achieved an overall accuracy of 86.2% ($\kappa = 0.554$), whereas the Krishna→Pichavaram transfer achieved only 48.9% ($\kappa = 0.196$). A same-domain baseline trained and tested within Pichavaram achieved over 98% accuracy, indicating that the poor cross-domain result is attributable to the magnitude of the domain gap rather than to the discriminative quality of the underlying features. These results demonstrate both the practical utility and the limitations of shallow, second-order statistical domain alignment for cross-region ecological mapping, and motivate the use of more expressive, non-linear or class-conditional adaptation strategies for regions with larger domain divergence.

Contents

Certificate ii
Declaration iii
Acknowledgements iv
Abstract v
Contents vi
1.1 NISAR Satellite Data 1
1.2 Unsupervised Domain Adaptation and Machine Learning Techniques 1
1.3 Study Areas 2
1.3.1 Krishna Mangroves (Source Domain) 2
1.3.2 Malad Mangroves (Target Domain 1) 2
1.3.3 Pichavaram Mangroves (Target Domain 2) 3
1.4 Cross-Domain Transfer Learning for Mangrove Species Classification 3
1.5 Organization of the Report 3
2.1 Domain Alignment 4
2.2 Correlation Alignment (CORAL) 4
2.3 Subspace Alignment 5
2.4 Hybrid CORAL–Subspace Alignment 6
2.4.1 CORAL followed by SA 6
2.4.2 SA followed by CORAL 7
2.5 Comparative Analysis 7
2.5.1 Stable Pixel Selection 8
2.5.2 SA Subspace Dimension Selection 9
2.5.3 Quantitative Comparison 11
2.5.4 Qualitative Visualisation 11
2.6 Chapter Summary 12
3.1 Proposed Framework 14
3.2 Dataset Preparation 14
3.3 Feature Representation 15
3.4 Domain Alignment using CORAL 15
3.5 One-vs-Rest Random Forest Classification 15
3.6 Probability Map Generation 16
3.7 Water Masking 16
3.8 Mangrove Species Map Generation 16
3.9 Accuracy Assessment 16

3.10 Chapter Summary 17
4.1 Cross-Domain Classification: Krishna → Study Area 1 (Malad) 18
4.2 Cross-Domain Classification: Krishna → Study Area 2 (Pichavaram) 19
4.2.1 Same-Domain Baseline for Pichavaram 21
4.3 Comparative Analysis 22
4.4 Discussion 23
4.4.1 Uses: Where CORAL-Based Domain Adaptation Succeeds 23
4.4.2 Limitations: Where CORAL-Based Domain Adaptation Fails 23
4.5 Chapter Summary 24
5.1 Conclusions 25
5.2 Contributions of the Study 25
5.3 Limitations 26
5.4 Future Scope 26
References viii
Appendices ix
Appendix A — CORAL Implementation ix
Appendix B — Class Distribution in the Krishna Ground-Truth Dataset ix

Chapter 1

Introduction

1.1 NISAR Satellite Data

The NASA–ISRO Synthetic Aperture Radar (NISAR) mission, launched on 30 July 2025 aboard a GSLV-F16 rocket from the Satish Dhawan Space Centre, is a joint Earth observation mission developed by NASA and the Indian Space Research Organisation (ISRO). NISAR is the first satellite to carry two synthetic aperture radar (SAR) instruments operating at different frequencies on a single platform: an L-band SAR (24 cm wavelength), contributed by NASA, and an S-band SAR (approximately 9–12 cm wavelength), contributed by ISRO. Operating from a sun-synchronous orbit at an altitude of approximately 747 km, the mission is designed to image nearly all of Earth's land and ice surfaces at least once every 12 days, providing all-weather, day-and-night coverage that is largely unaffected by cloud cover — a significant advantage over optical remote sensing for regions such as tropical and sub-tropical mangrove forests, which are frequently obscured by cloud and haze.

The lower-frequency L-band signal penetrates vegetation canopies more effectively than shorter-wavelength radar or optical sensors, making it particularly well suited to characterising forest structure, biomass, and wetland inundation — all of which are directly relevant to mapping mangrove extent, density, and species-level composition. The broader motivation for this project is to work towards leveraging SAR-derived information of this kind for mangrove species mapping. In the present study, the classification and domain adaptation pipeline was developed and evaluated using pre-processed 64-band feature embeddings supplied as part of the internship project for the Krishna, Malad, and Pichavaram mangrove regions, rather than by processing raw sensor imagery directly; the methodological contribution of this work lies in the domain adaptation and classification stages built on top of these features, described in Chapters 2 to 4.

1.2 Unsupervised Domain Adaptation and Machine Learning Techniques

A classifier trained on labelled data from one geographic region (the source domain) commonly suffers a drop in accuracy when applied directly to another region (the target domain) for which no labels are available. This effect, referred to as domain shift, arises because the source and target domains, although describing conceptually similar classes, follow different underlying feature distributions — owing to differences in species composition, canopy density, tidal and sediment conditions, or acquisition characteristics between regions. Unsupervised domain adaptation (UDA)

is the family of techniques concerned with transferring a classifier learned on a labelled source domain to an unlabelled target domain by explicitly reducing this distributional mismatch.

UDA approaches are broadly grouped into sample re-weighting methods, model-based methods that impose transfer constraints on model parameters, and subspace or feature alignment methods that seek a shared, domain-invariant representation onto which both domains can be projected. This study focuses on the last category — specifically Correlation Alignment (CORAL) and Subspace Alignment (SA), which respectively align the second-order statistics (covariance) and the principal subspaces of the source and target features using simple, closed-form linear transformations. These shallow methods are computationally inexpensive, require no target labels, and are effective when the domain shift is primarily statistical rather than deeply non-linear — as contrasted with deep, adversarial domain-adaptation approaches that learn a non-linear invariant representation end-to-end but at substantially greater computational and data cost. Following domain alignment, a one-vs-rest Random Forest classifier, trained on labelled source-domain samples, is used to assign species-level class probabilities to every pixel of the aligned target domain.

1.3 Study Areas

1.3.1 Krishna Mangroves (Source Domain)

The Krishna river delta, on the east coast of Andhra Pradesh, hosts one of the largest contiguous mangrove ecosystems on the Indian east coast. The delta supports a mosaic of true mangrove species — predominantly Avicennia marina at varying canopy densities (dense, moderate, and sparse), together with Excoecaria agallocha, Bruguiera gymnorhiza, and Rhizophora species — interspersed with associated halophytic vegetation (e.g. Suaeda and Prosopis), tidal creeks, mudflats, and sand bars. A field-verified ground-truth shapefile covering 14 land-cover and species classes over the Krishna delta was used as the labelled source domain for this study.

1.3.2 Malad Mangroves (Target Domain 1)

The Malad creek mangrove belt, located along the western coast within the Mumbai metropolitan region of Maharashtra, represents a distinctly different ecological and urban setting from the Krishna delta. Malad's mangrove stands are dominated by Avicennia marina and Rhizophora, but are subject to substantially greater anthropogenic pressure, tidal flushing characteristics, and surrounding land-use context than the relatively undisturbed Krishna delta, making it a moderately divergent cross-region target domain.

1.3.3 Pichavaram Mangroves (Target Domain 2)

The Pichavaram mangrove forest, situated between the Vellar and Coleroon river estuaries on the Tamil Nadu coast, is one of the most extensively studied mangrove wetlands in India. It is characterised by a dense network of tidal creeks and Rhizophora- and Avicennia-dominated stands. Owing to differences in latitude, estuarine hydrology, and canopy structure relative to the Krishna delta, Pichavaram represents a considerably larger domain shift from the source domain than Malad, a distinction that is borne out quantitatively in the cross-domain classification results presented in Chapter 4.

1.4 Cross-Domain Transfer Learning for Mangrove Species Classification

The central problem addressed in this report is as follows: given a Random Forest classifier trained on labelled feature embeddings from the Krishna delta, how can this classifier be adapted to reliably classify mangrove species in the Malad and Pichavaram regions, for which no ground-truth labels are available? Applying the Krishna-trained classifier directly to these target regions, without any adaptation, is expected to perform poorly because the feature distributions of the same nominal species class differ across regions. The approach adopted in this study is to first identify the most effective shallow domain alignment technique on a controlled, within-region temporal domain-shift problem (Krishna 2019 versus Krishna 2024 imagery), and then to apply the selected technique — CORAL — to the substantially harder cross-region transfer task of classifying Malad and Pichavaram using only Krishna-derived training labels.

1.5 Organization of the Report

Chapter 2 introduces domain alignment techniques — Subspace Alignment, Correlation Alignment, and their sequential hybrids — and presents a comparative evaluation of these methods on the Krishna 2019→2024 temporal domain-shift task, leading to the selection of CORAL as the adaptation method used in the rest of the study. Chapter 3 describes the proposed cross-domain mangrove species classification framework, covering dataset preparation, CORAL-based domain alignment, one-vs-rest Random Forest classification, probability map generation, water masking, and accuracy assessment. Chapter 4 presents the cross-domain classification results for the Krishna→Malad and Krishna→Pichavaram transfer tasks, along with a comparative analysis and discussion of the uses and limitations of the adopted approach. Chapter 5 concludes the report and outlines the contributions, limitations, and scope for future work.

Chapter 2

Domain Alignment Techniques

2.1 Domain Alignment

When a source domain S and a target domain T describe conceptually the same set of classes but are drawn from different underlying feature distributions, a classifier trained purely on S cannot be assumed to generalise to T. Domain alignment techniques address this by explicitly transforming the feature representation of one or both domains so that their distributions become statistically or geometrically closer, thereby making a classifier trained on the aligned source domain more reliable when applied to the target domain. The principal use of domain alignment in this study is to make it possible to train a mangrove species classifier using only the labelled Krishna data and then apply it, with reasonable confidence, to unlabelled regions (Malad and Pichavaram) for which collecting comparable ground truth would otherwise be costly and time-consuming. This is of direct practical value for large-scale mangrove monitoring, where field verification is not feasible for every region of interest.

At the same time, domain alignment methods carry an important limitation: they operate without access to target-domain labels, and therefore have no direct means of verifying whether the aligned feature space preserves class-discriminative information, or whether it merely makes the two domains look similar in aggregate. Shallow, statistics-based alignment methods such as Subspace Alignment and Correlation Alignment are attractive for their simplicity, closed-form solutions, and low computational cost, but they implicitly assume that a linear transformation is sufficient to correct the domain shift. This assumption becomes increasingly untenable as the magnitude and non-linearity of the domain shift grows — a limitation this study returns to and quantifies directly in Chapter 4.

2.2 Correlation Alignment (CORAL)

Correlation Alignment (CORAL) aligns the second-order statistics — i.e. the feature covariance — of the source and target domains through a closed-form linear transformation, without requiring any target labels. Given source features X_S and target features X_T, the class-wise covariance matrices are computed as:

$$C_S = \frac{1}{n-1}(X_S - \bar{X}_S)^T(X_S - \bar{X}_S), \quad C_T = \frac{1}{n-1}(X_T - \bar{X}_T)^T(X_T - \bar{X}_T)$$

CORAL then seeks a transformation that whitens the source features with respect to their own covariance and re-colours them with the target covariance, so that the second-order statistics of the transformed source features match those of the target domain:

$$Class(x) = \arg \max_{c \in \{1,\dots,C\}} P_c(x)$$

which admits the following closed-form solution:

$$X_{aligned} = X_S \, C_S^{-1/2} \, C_T^{1/2}$$

where the exponents denote the matrix square root and inverse square root computed via eigendecomposition of the (symmetric, positive semi-definite) covariance matrices. In practice, a small ridge term is added to each covariance matrix before inversion to ensure numerical stability:

$$\hat{C} = C + \epsilon I, \qquad \epsilon = 10^{-5}$$

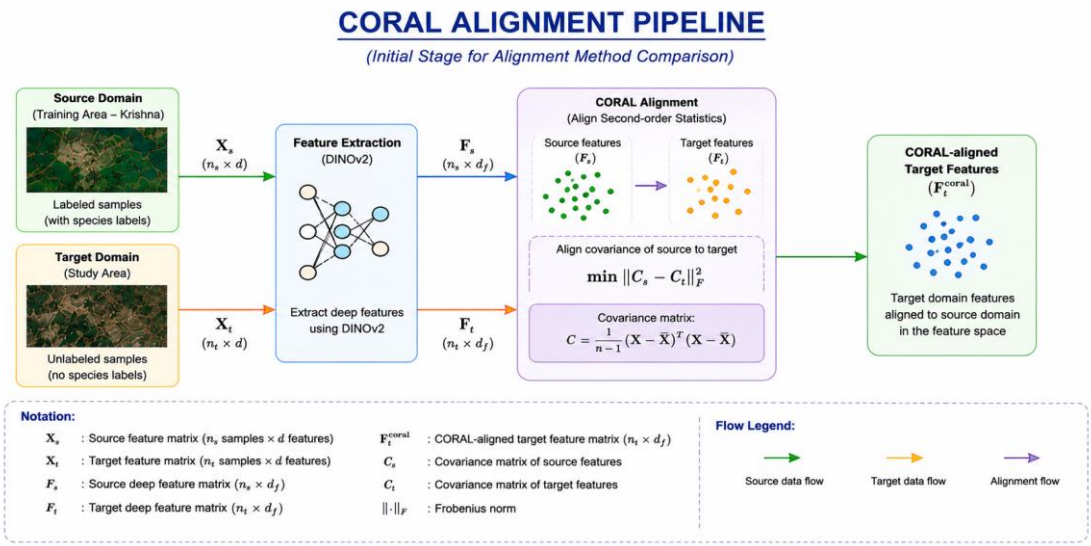


Figure 2.1: CORAL alignment pipeline used as the initial stage for domain alignment method comparison.

2.3 Subspace Alignment

Subspace Alignment (SA), following the formulation of Fernando et al. (2014), represents the source and target domains by the subspaces spanned by their respective top-d principal components, obtained independently via Principal Component Analysis (PCA):

$$X_S = PCA(S, d), \qquad X_T = PCA(T, d)$$

SA then learns a linear transformation matrix M that maps the source subspace basis onto the target subspace basis by minimising the Frobenius-norm (Bregman) divergence between the transformed source basis and the target basis. Because both bases are orthonormal, this optimisation reduces to a simple closed-form expression:

$$M^* = \arg\min_M \|X_S M - X_T\|_F^2 = X_S^T X_T$$

The target-aligned source coordinate system is then obtained as:

$$X_a = X_S X_S^T X_T$$

Source samples are subsequently projected through X_a and target samples through X_T, placing both domains in a common d-dimensional aligned subspace in which a classifier trained on the source domain can be applied to the target domain.

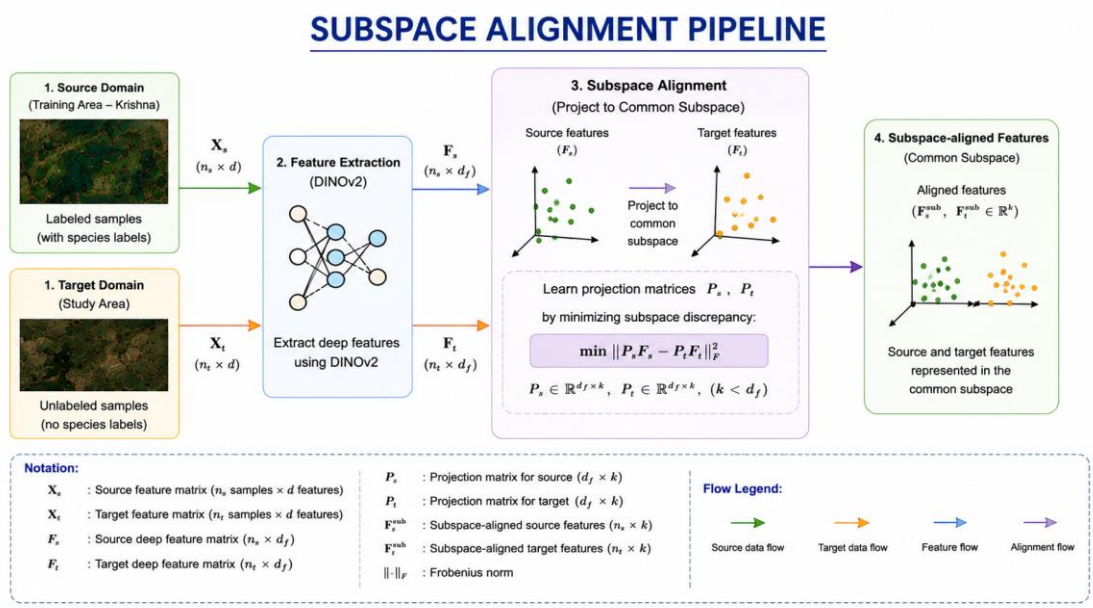


Figure 2.2: Subspace Alignment (SA) pipeline.

2.4 Hybrid CORAL–Subspace Alignment

In addition to CORAL and SA individually, two sequential hybrid strategies were evaluated to determine whether combining statistical (covariance-based) and geometric (subspace-based) alignment could outperform either method alone.

2.4.1 CORAL followed by SA

In the CORAL→SA strategy, CORAL is first applied to match the source and target covariance matrices; Subspace Alignment is then applied to the CORAL-transformed source features to further reduce any residual geometric (subspace) discrepancy:

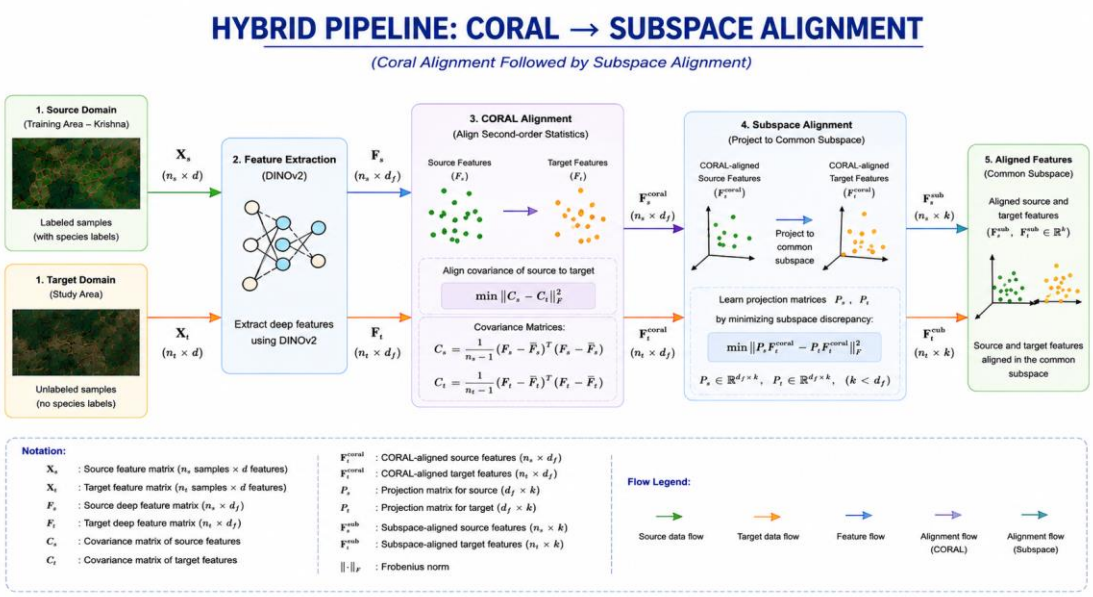


Figure 2.3: Hybrid CORAL → SA alignment pipeline.

2.4.2 SA followed by CORAL

In the SA→CORAL strategy, the order is reversed: Subspace Alignment is applied first to reduce the geometric discrepancy between the source and target subspaces, and CORAL is subsequently applied to align the covariance statistics of the SA-transformed features with those of the target domain:

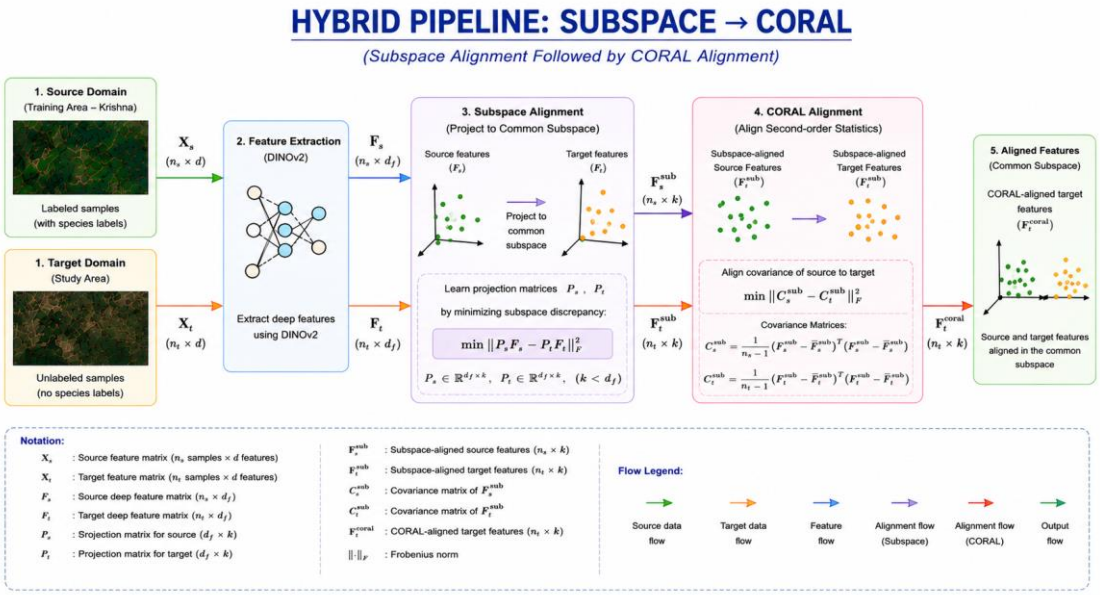


Figure 2.4: Hybrid SA → CORAL alignment pipeline.

2.5 Comparative Analysis

To identify the most effective domain alignment technique before applying it to the harder cross-region transfer task, all four methods above (SA, CORAL, CORAL→SA, and SA→CORAL) were

benchmarked on a controlled, within-region temporal domain-shift problem: the 64-band Krishna delta feature embeddings from 2019 (source) versus 2024 (target). This temporal shift is considerably smaller than the cross-region shift addressed in Chapter 4, and therefore serves as a useful controlled setting in which to compare alignment methods before committing to one for the harder task.

2.5.1 Stable Pixel Selection

Not every pixel is equally informative for estimating the domain-alignment transformation — pixels that have changed substantially in feature space between 2019 and 2024 (e.g. due to genuine land-cover change) would bias the estimated covariance and subspace statistics if included indiscriminately. To address this, a stable-pixel selection step, used specifically to prepare input for the CORAL transformation, was applied. For every pixel, the cosine distance between its 2019 and 2024 feature vectors was computed:

$$d_{\cos}(x_i) = 1 - \frac{x_i^{(2019)} \cdot x_i^{(2024)}}{\left\|x_i^{(2019)}\right\| \left\|x_i^{(2024)}\right\|}$$

Pixels with the smallest cosine distance are the most temporally stable and were retained for estimating the domain-alignment statistics.

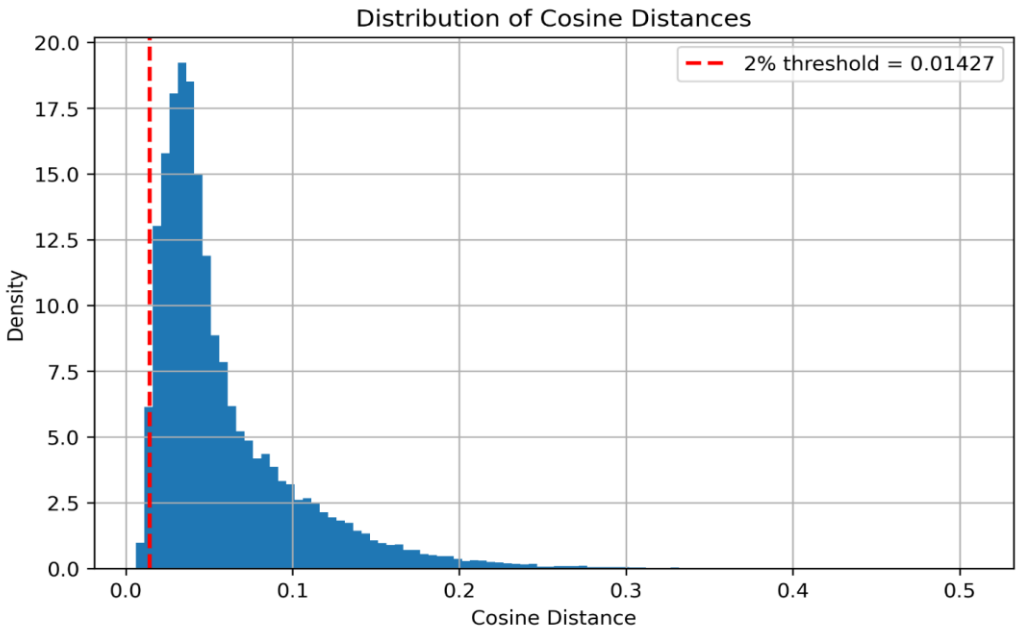


Figure 2.6: Distribution of cosine distances, with the 2% threshold (0.01427) marked.

A 1% threshold gives the highest similarity but an insufficient number of stable pixels for reliable statistics estimation, while a 5% threshold admits a visibly larger proportion of unstable pixels. A

2% threshold was therefore adopted as a practical compromise, yielding 41,937 stable pixels used to estimate the CORAL transformation.

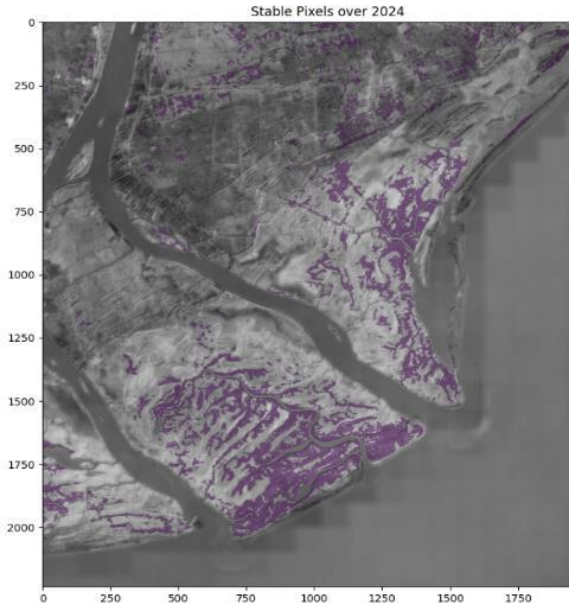


Figure 2.7: Spatial distribution of stable pixels (highlighted) overlaid on the 2024 Krishna image

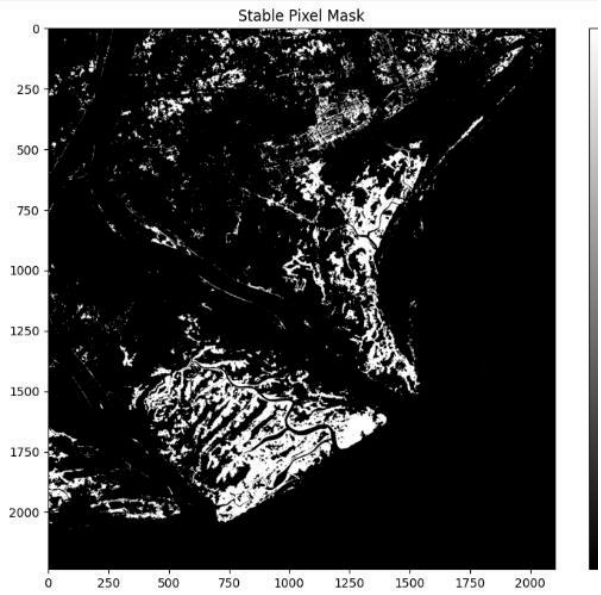


Figure 2.8: Binary stable-pixel mask.

2.5.2 SA Subspace Dimension Selection

Subspace Alignment requires a choice of subspace dimension d, the number of principal components used to represent each domain. Too small a value of d discards useful spectral information; too large a value retains redundant, noise-dominated components that do not aid alignment. The covariance gap and subspace distance were evaluated for every value of d from 1 to 64 (Figure 2.9). The subspace distance decreases progressively with increasing d, whereas the covariance gap follows the opposite trend beyond a modest dimensionality, indicating that increasing d does not uniformly improve statistical alignment.

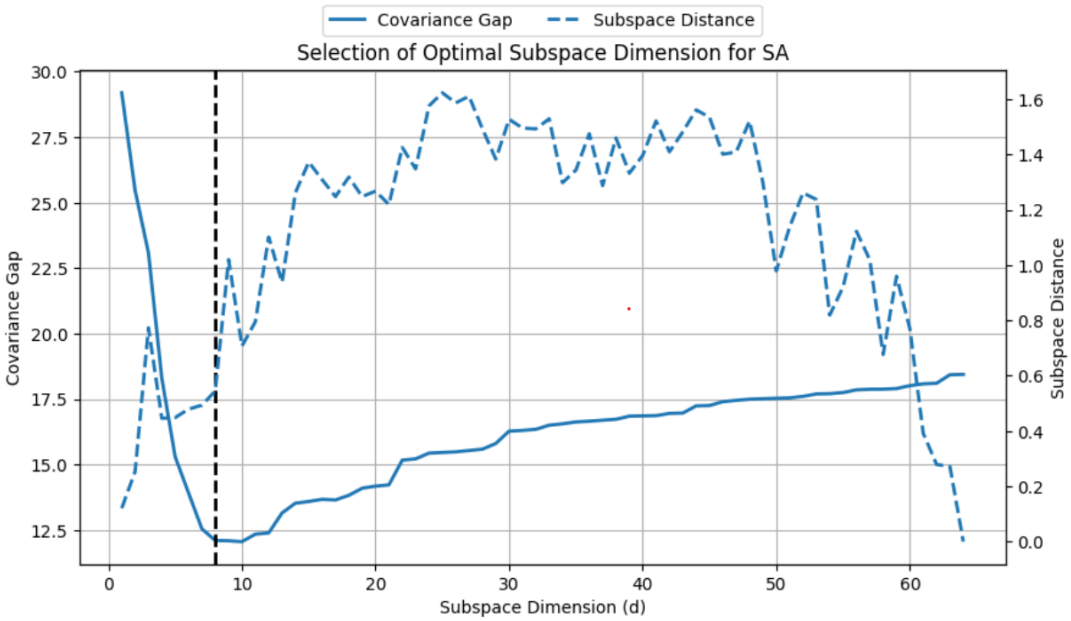


Figure 2.9: Covariance gap and subspace distance as a function of SA subspace dimension d.

A Pareto analysis of covariance gap against subspace distance across all values of d (Figure 2.10) was used to identify the dimension offering the best joint compromise between the two metrics. Dimension d = 8 was selected, as it lies close to the Pareto-optimal region where both metrics are simultaneously low relative to neighbouring dimensions; this value was used for all subsequent Subspace Alignment experiments.

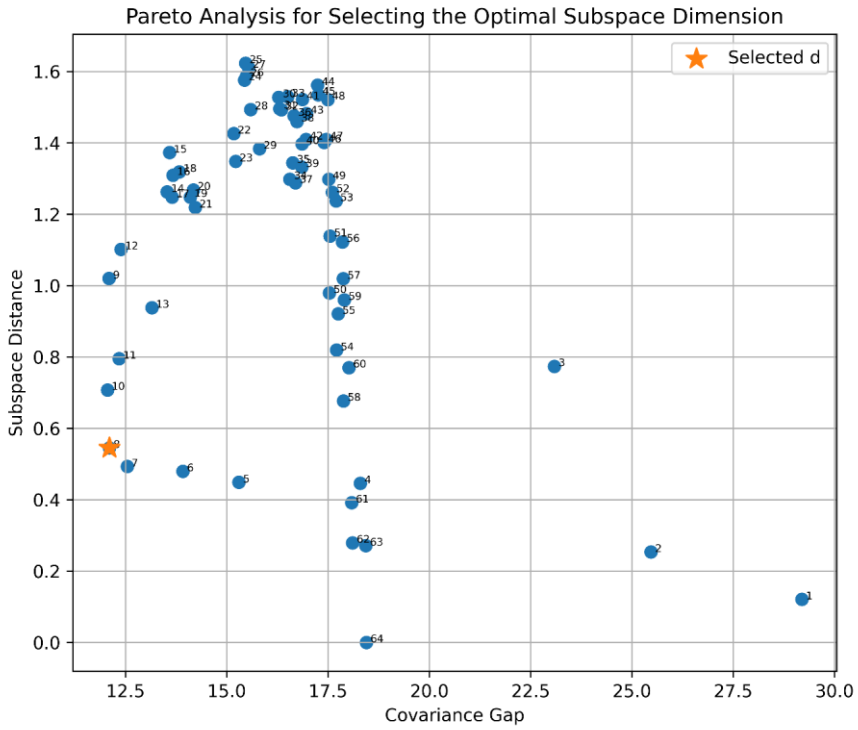


Figure 2.10: Pareto analysis for selecting the optimal SA subspace dimension (selected d = 8).

2.5.3 Quantitative Comparison

The initial covariance gap and subspace distance between the 2019 and 2024 Krishna feature embeddings, prior to any adaptation, were 18.443 and 6.184 respectively. Table 2.1 summarises the covariance gap and subspace distance achieved by each alignment method, before and after adaptation.

Method	Cov. Gap (Before)	Cov. Gap (After)	Subspace Dist. (Before)	Subspace Dist. (After)
Original (No Adaptation)	18.443	18.443	6.184	6.184
Subspace Alignment (SA)	18.443	12.100	6.184	0.545
SA + CORAL	18.443	5.415	6.184	0.545
CORAL + SA	18.443	3.902	6.184	0.00017
CORAL	18.443	0.00175	6.184	0.00027

Table 2.1: Covariance gap and subspace distance before and after each domain alignment method (Krishna 2019 → 2024).

The comparison shows that CORAL achieved the strongest reduction in both the covariance gap (from 18.443 to 0.00175, a reduction of over 99.9%) and the subspace distance (from 6.184 to 0.00027), outperforming SA and both hybrid variants. SA alone substantially reduced the subspace distance but left a comparatively large residual covariance gap (12.100). The two hybrid strategies improved one metric at the expense of the other relative to CORAL alone: CORAL→SA achieved a very low subspace distance (0.00017) but a markedly higher covariance gap (3.902) than CORAL alone, while SA→CORAL achieved an intermediate covariance gap (5.415) without improving on CORAL's subspace distance. This indicates that, for this dataset, applying CORAL alone — rather than combining it sequentially with SA in either order — gives the strongest joint statistical and geometric alignment.

2.5.4 Qualitative Visualisation

To visually assess the effect of CORAL alignment, the stable-pixel samples were reduced from 64 to 20 dimensions using PCA and then embedded into two dimensions using a single t-distributed Stochastic Neighbour Embedding (t-SNE) model, so that the source, target, and CORAL-aligned samples share a common embedding space. As shown in Figure 2.11, the 2019 and 2024 samples already overlap substantially before adaptation, with several localised regions of mismatch; after CORAL alignment, these localised discrepancies visibly reduce while the overall structure of the feature space is preserved.

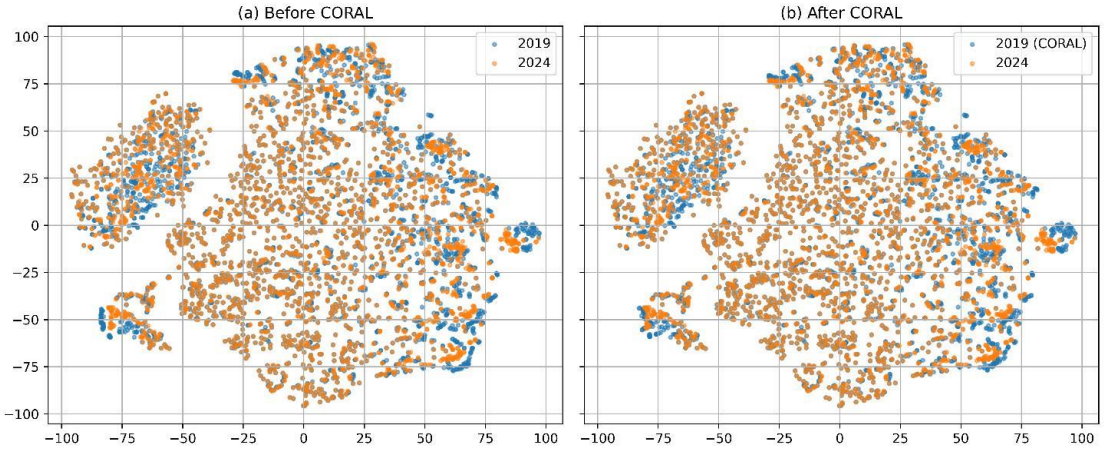


Figure 2.11: t-SNE visualisation of source (2019) and target (2024) samples before and after CORAL alignment.

The covariance difference matrices in Figure 2.12 provide a complementary, band-wise view of the same effect: before adaptation, several spectral-band pairs exhibit high-intensity absolute covariance differences; after CORAL alignment, the difference matrix becomes almost uniformly low in intensity, consistent with the near-total reduction in the quantitative covariance gap reported in Table 2.1.

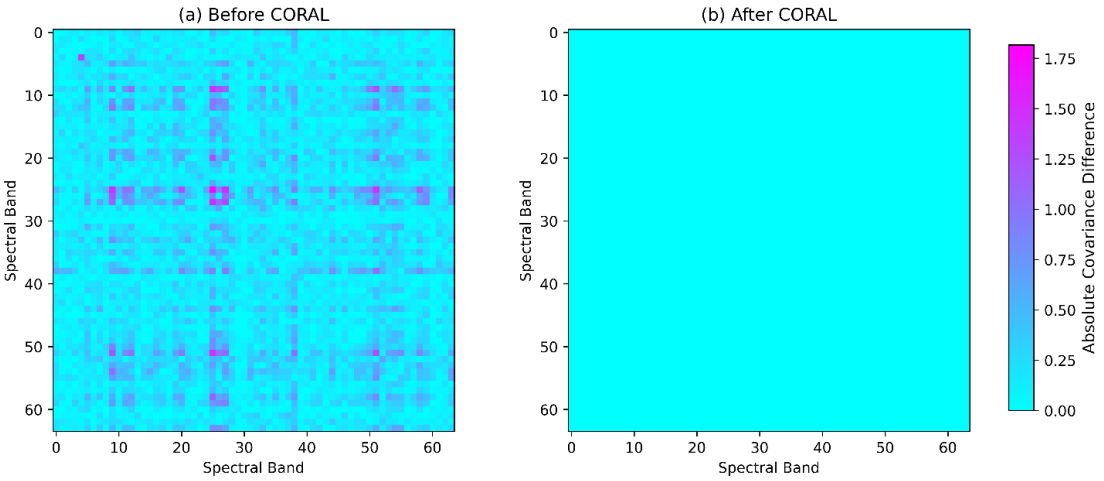


Figure 2.12: Absolute covariance-difference matrices between source and target features, before and after CORAL alignment.

On the basis of this combined quantitative and qualitative evidence, CORAL was selected as the domain alignment method for the cross-region mangrove species classification task described in Chapters 3 and 4.

2.6 Chapter Summary

This chapter introduced the mathematical formulation of Subspace Alignment, Correlation Alignment, and their two sequential hybrid combinations, and compared all four methods on a

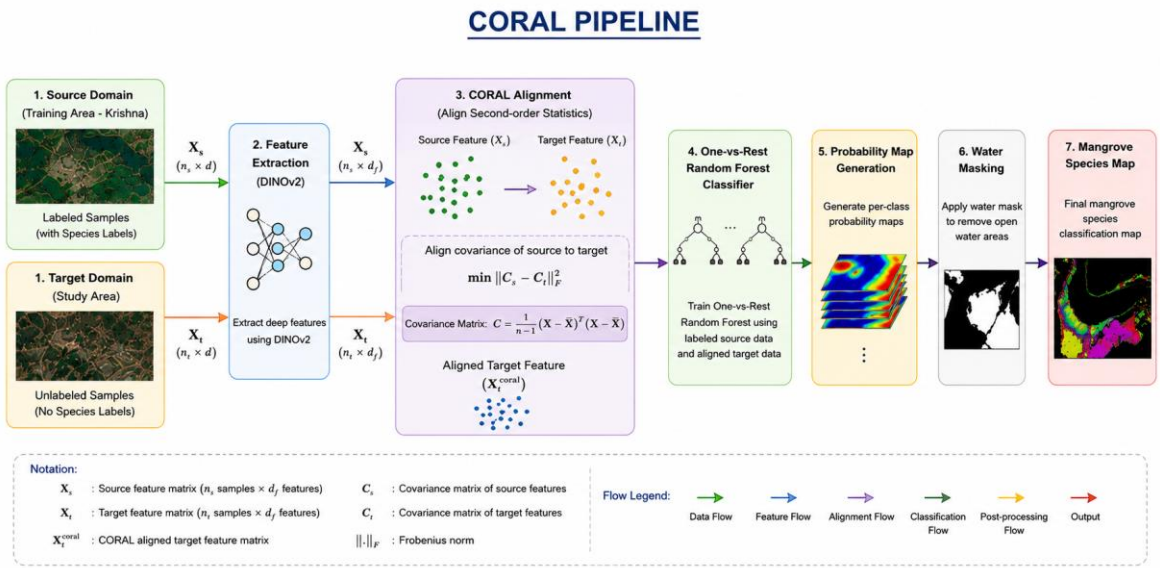
controlled Krishna 2019→2024 temporal domain-shift benchmark using covariance gap and subspace distance as quantitative alignment metrics. CORAL achieved the strongest joint reduction in both metrics and was selected as the domain alignment method used in the remainder of this study. The next chapter describes how CORAL is incorporated into a complete cross-domain mangrove species classification framework.

Chapter 3

Proposed Cross-Domain Mangrove Species Classification Framework

3.1 Proposed Framework

Building on the domain alignment comparison of Chapter 2, this chapter describes the complete pipeline used to classify mangrove species in the Malad and Pichavaram target domains using only labelled Krishna delta source data. The framework consists of six stages: (i) dataset preparation and class-schema alignment across regions; (ii) CORAL-based domain alignment of the target feature embeddings to the Krishna source distribution; (iii) training of a one-vs-rest Random Forest classifier on Krishna ground-truth samples; (iv) generation of per-species probability maps over the aligned target domain; (v) water masking to exclude non-vegetated tidal and creek areas from the species map; and (vi) generation of the final mangrove species map and accuracy assessment. Each stage is described in the sections that follow.



3.2 Dataset Preparation

The source domain consists of 64-band pre-processed feature embeddings covering the Krishna delta, together with a field-verified ground-truth shapefile containing 14 classes (mangrove species at varying canopy density, associated vegetation, and non-vegetated cover types such as creek water, mudflat, and sand bars). The shapefile was rasterised onto the Krishna feature grid, and only the classes relevant to the target domains' class schema were retained for training. For the Malad

transfer task, three classes were used — Avicennia marina (dense), Rhizophora, and waterbody — while for the Pichavaram transfer task, four classes were used, distinguishing Avicennia marina at dense, moderate, and sparse canopy density in addition to Rhizophora. Because the Krishna ground-truth class codes do not correspond directly to the class schema used in each target region, an explicit class-remapping was applied — for example, mapping Krishna class 11 (Rhizophora_sp) to class 1 in the Pichavaram schema, and Krishna classes 1–3 (A. marina dense / moderate / sparse) to classes 2–4 respectively.

3.3 Feature Representation

Both the source (Krishna) and target (Malad, Pichavaram) feature rasters consist of 64-band, pixel-wise feature embeddings, supplied as pre-processed input as part of the internship dataset. These embeddings form the feature space $X \in R^{64}$ on which all subsequent domain alignment and classification in this study is performed; no additional feature-extraction step was carried out as part of this work.

3.4 Domain Alignment using CORAL

Following the formulation and comparative justification given in Chapter 2, the target domain feature embeddings (Malad or Pichavaram) were aligned to the Krishna source distribution using CORAL. Concretely, the Krishna source features were treated as X_S and the target-region features as X_T; the CORAL whitening–recolouring transform (Section 2.2) was applied to the source features so that their covariance structure matches that of the target region, after which the Krishna-trained classifier operates directly on the CORAL-aligned feature space. The implementation followed the closed-form solution of Section 2.2, computing the matrix square root and inverse square root of the source and target covariance matrices via eigendecomposition, with a ridge term of 1×10^{-5} added to each covariance matrix prior to inversion to avoid numerical instability arising from near-singular covariance estimates.

3.5 One-vs-Rest Random Forest Classification

A separate binary Random Forest classifier was trained for each mangrove species class using a one-vs-rest strategy: for a given class, all Krishna training pixels belonging to that class were treated as the positive class and all remaining training pixels as the negative class. This strategy was adopted, in preference to a single multi-class classifier, because it allows an independent probability surface to be generated for each species (Section 3.6) and accommodates the differing class schemas used across the two target regions. Two Random Forest configurations were used at different stages of this study: the primary cross-domain classification experiments (Chapter 4) used 500 estimators, the square-root heuristic for the number of features considered at each split, and

balanced class weighting to counteract the class imbalance visible in the Krishna ground-truth distribution (Section 3.2); an additional same-domain baseline classifier used for comparison in Section 4.2 (trained and tested within Pichavaram) used 200 estimators with a maximum tree depth of 20.

3.6 Probability Map Generation

For each species, the corresponding one-vs-rest Random Forest classifier was applied to every pixel of the CORAL-aligned target-domain feature raster, producing a continuous per-pixel probability surface indicating the estimated likelihood that the pixel belongs to that species. These probability maps (see Figures 4.1 and 4.4) provide a more informative and interpretable intermediate output than a hard classification alone, since they expose the classifier's relative confidence across species and across the spatial extent of the target region.

3.7 Water Masking

Because Random Forest probability surfaces are computed independently for each species and are not mutually exclusive, tidal creeks, open water, and other non-vegetated surfaces can occasionally receive spuriously elevated probability values for vegetated species classes. A water class was therefore included explicitly among the one-vs-rest classifiers, and pixels most strongly associated with the water class were masked out prior to final species-map generation, ensuring that only vegetated pixels are assigned a mangrove species label in the final map.

3.8 Mangrove Species Map Generation

The final categorical mangrove species map for each target region was generated by assigning every non-water pixel to the species class with the highest one-vs-rest probability at that location (an arg-max decision rule across the per-species probability surfaces produced in Section 3.6). The resulting species maps for Malad and Pichavaram are presented in Sections 4.1 and 4.2 respectively.

3.9 Accuracy Assessment

The classification accuracy of the cross-domain species maps was assessed against available target-domain ground-truth samples using overall accuracy (OA), the Cohen's kappa coefficient (κ), and the full confusion matrix and per-class classification report (precision, recall, and F1-score). Overall accuracy is defined as:

$$OA = \frac{\sum_{i=1}^{C} n_{ii}}{N} \times 100\%$$

where n_ii is the number of correctly classified samples for class i, k is the number of classes, and N is the total number of test samples. Cohen's kappa additionally corrects for the accuracy expected by chance agreement, P_e:

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

and is generally regarded as a more conservative and reliable indicator of classification performance than overall accuracy alone, particularly under class imbalance.

3.10 Chapter Summary

This chapter described the full cross-domain mangrove species classification framework: dataset preparation and class-schema alignment, CORAL-based domain alignment of the target feature space to the Krishna source distribution, one-vs-rest Random Forest classification, per-species probability map generation, water masking, final species-map generation, and accuracy assessment methodology. Chapter 4 applies this framework to the Malad and Pichavaram target domains and presents the resulting classification accuracies, confusion matrices, and species maps.

Chapter 4

Results and Discussion

4.1 Cross-Domain Classification: Krishna → Study Area 1 (Malad)

The CORAL-aligned, Krishna-trained one-vs-rest Random Forest classifier was first applied to the Malad creek mangrove region, using a three-class schema (Avicennia marina dense, Rhizophora, and waterbody). Figure 4.1 shows the per-species probability maps produced by the classifier across the Malad study area.

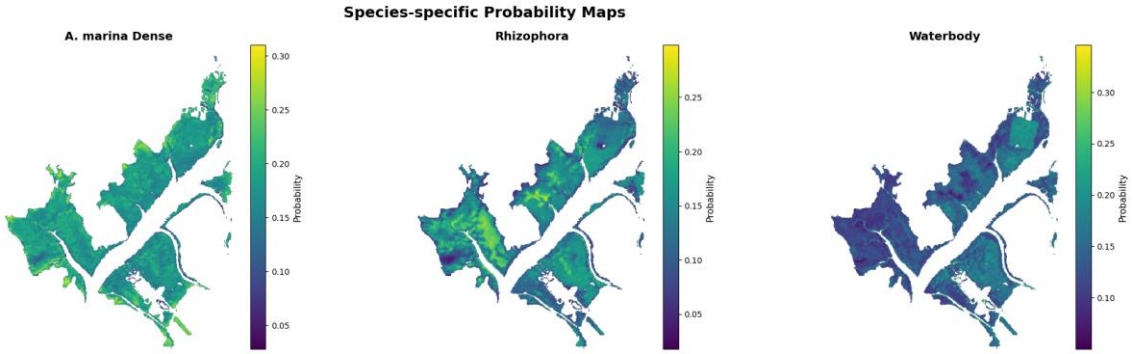


Figure 4.1: Species-specific probability maps for the Krishna → Malad cross-domain classification (A. marina dense, Rhizophora, Waterbody).

The final mangrove species map, obtained by assigning each non-water pixel to the species with the highest probability, is shown in Figure 4.3, together with the corresponding confusion matrix against available Malad ground-truth samples in Figure 4.2.

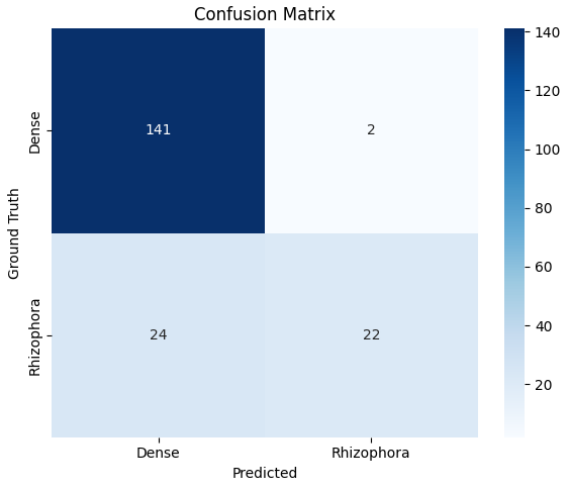


Figure 4.2: Confusion matrix for the Krishna → Malad cross-domain classification

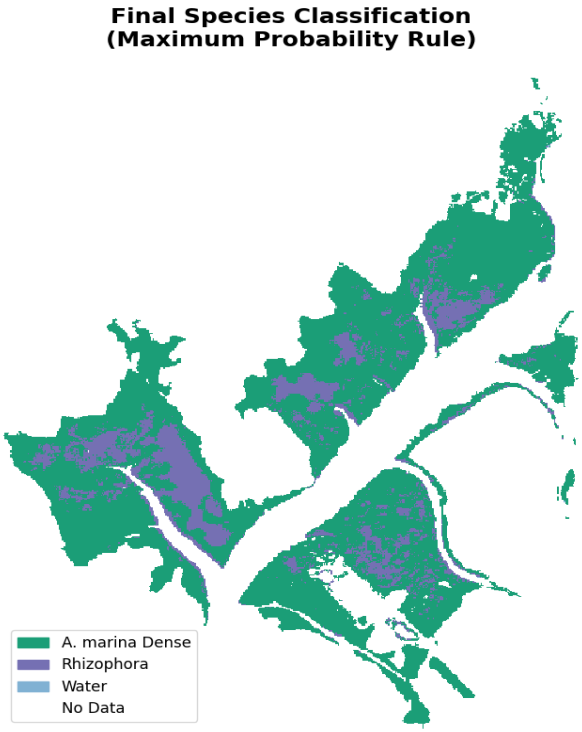


Figure 4.3: Final CORAL-aligned mangrove species classification map for Malad.

The Krishna → Malad transfer achieved an overall accuracy of 86.2% and a Cohen's kappa of 0.554. The comparatively high overall accuracy, together with a moderate kappa value, indicates that the CORAL-aligned classifier is able to correctly identify the dominant species classes in Malad for a large majority of pixels, while some confusion remains between classes with more subtle spectral or structural differences, as detailed in the confusion matrix of Figure 4.3.

4.2 Cross-Domain Classification: Krishna → Study Area 2 (Pichavaram)

The same CORAL-aligned, Krishna-trained classifier was next applied to the Pichavaram mangrove forest, using a four-class schema distinguishing Avicennia marina at dense, moderate, and sparse canopy density, together with Rhizophora. Figure 4.4 shows a subset of the resulting per-species probability maps.

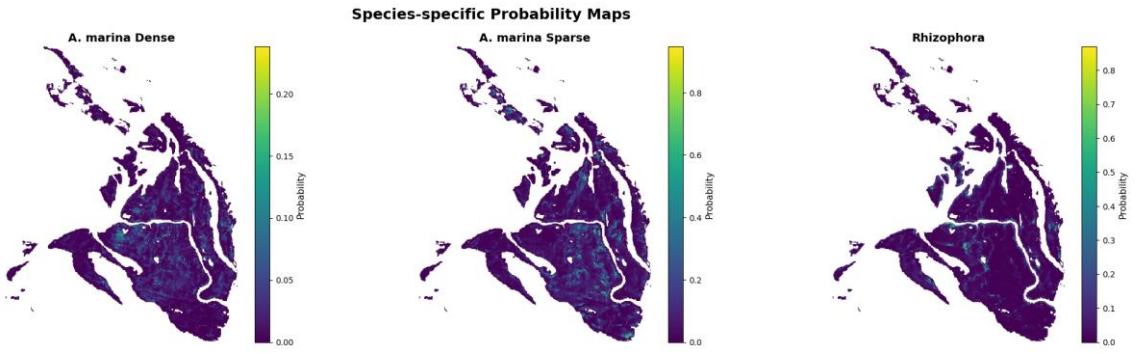


Figure 4.4: Species-specific probability maps for the Krishna → Pichavaram cross-domain classification.

Figure 4.5 shows the mean probability assigned by each Krishna-trained species model to the true species groups present in the Pichavaram ground truth, aggregated over all test pixels of each true class. The matrix is close to uniformly low across most class pairs, with the Rhizophora model showing the highest mean self-probability (0.12) on true Rhizophora pixels, while the three Avicennia marina density classes show diffuse, generally low probabilities that are concentrated disproportionately on the sparse-canopy model rather than on their own respective density class — indicating that the Krishna-trained density-level distinctions transfer only weakly to Pichavaram.

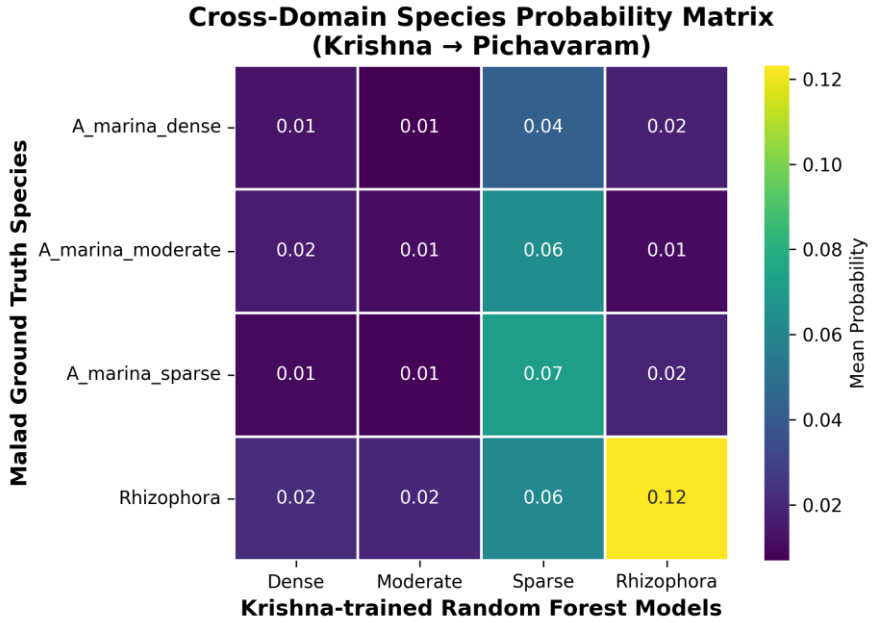


Figure 4.5: Cross-domain mean species probability matrix (Krishna-trained models × true Pichavaram species).

The confusion matrix and classification report for the Krishna → Pichavaram transfer are shown in Figure 4.6.

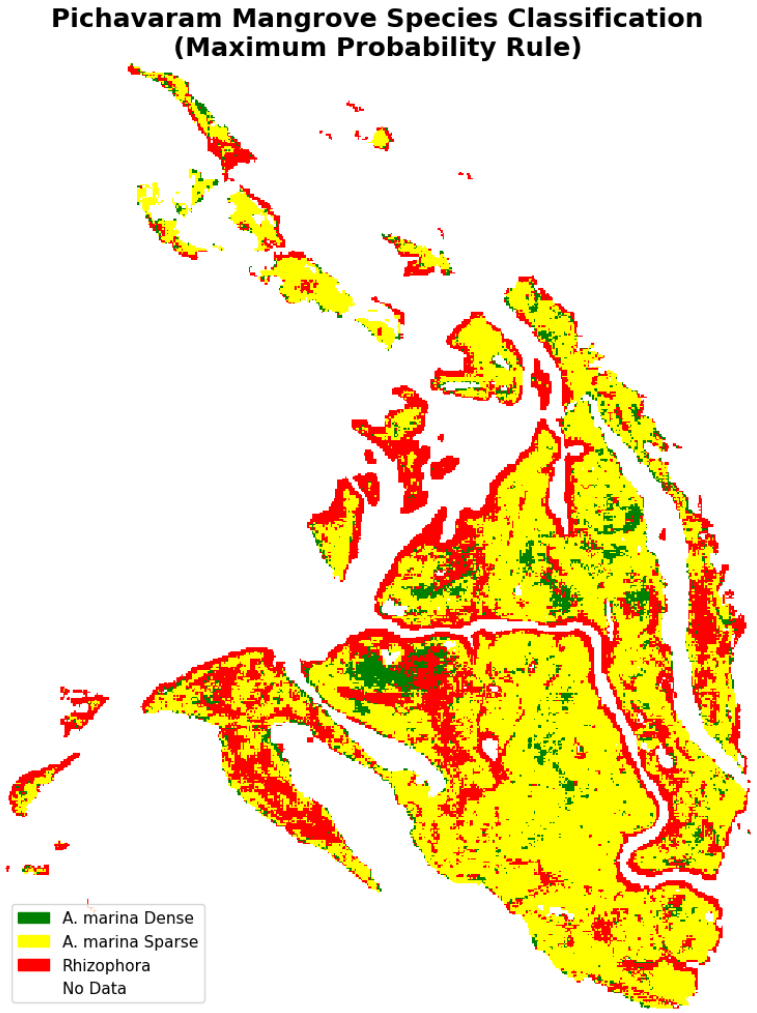


Figure 4.6: Confusion matrix for the Krishna → Pichavaram cross-domain classification.

The Krishna → Pichavaram transfer achieved an overall accuracy of only 48.9% and a Cohen's kappa of 0.196 — substantially lower than the Krishna → Malad result, and, given the four-class schema, only moderately better than chance-level performance.

4.2.1 Same-Domain Baseline for Pichavaram

To determine whether this poor performance stemmed from the CORAL-aligned features being inherently non-discriminative in Pichavaram, or from the cross-domain transfer itself, a same-domain baseline classifier was additionally trained and evaluated entirely within Pichavaram (i.e. trained and tested on Pichavaram samples, with no domain shift and no CORAL alignment involved). This baseline, using a Random Forest with 200 estimators and a maximum depth of 20, achieved an overall accuracy of approximately 98–99% and a kappa of approximately 0.99 (Figure 4.7).

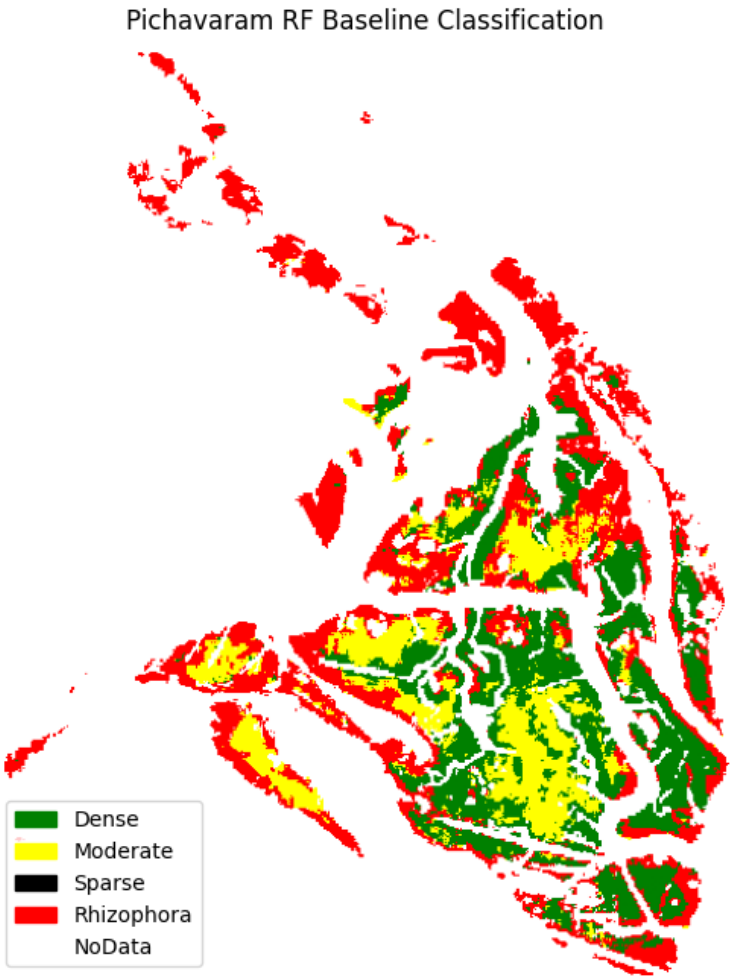


Figure 4.7: Confusion matrix for the same-domain Pichavaram baseline classifier (trained and tested within Pichavaram; no cross-domain transfer).

This near-ceiling same-domain accuracy demonstrates that the 64-band feature embeddings for Pichavaram are, in principle, highly discriminative for the species classes considered — the poor cross-domain result of Section 4.2 is therefore attributable to the magnitude of the Krishna→Pichavaram domain gap and the limitations of a single global, linear CORAL transformation in bridging it, rather than to any inherent lack of separability in the underlying feature space.

4.3 Comparative Analysis

Metric	Krishna → Malad	Krishna → Pichavaram	Pichavaram Same-Domain Baseline
Overall Accuracy	86.2%	48.9%	~98–99%
Cohen's Kappa (κ)	0.554	0.196	~0.99
Number of classes	3	4	4
Domain alignment used	CORAL	CORAL	None (same domain)

Table 4.1: Comparative summary of cross-domain and same-domain classification performance.

The contrast in Table 4.1 illustrates the central finding of this study: CORAL-based domain alignment enables a Krishna-trained classifier to transfer successfully to a moderately divergent target region (Malad), but its effectiveness degrades sharply as the magnitude of the domain shift increases (Pichavaram), even though the underlying target-domain features remain highly separable, as demonstrated by the same-domain baseline.

4.4 Discussion

4.4.1 Uses: Where CORAL-Based Domain Adaptation Succeeds

The Krishna → Malad result demonstrates the practical value of CORAL-based domain adaptation for cross-region mangrove mapping: an accuracy of 86.2% was achieved without any labelled data from Malad, using only a classifier trained on the Krishna delta and a single closed-form covariance-alignment transformation. This confirms that, when the domain shift between source and target regions is moderate, second-order statistical alignment is sufficient to bring the source-trained decision boundaries into good agreement with the target feature distribution — consistent with the near-total reduction in covariance gap achieved by CORAL in the controlled Krishna 2019→2024 benchmark of Chapter 2.

4.4.2 Limitations: Where CORAL-Based Domain Adaptation Fails

The Krishna → Pichavaram result exposes a clear limitation of CORAL, and of shallow, statistics-based domain alignment more generally. CORAL aligns only the global, second-order (covariance) statistics of the source and target domains using a single linear transformation; it does not model class-conditional shift (i.e. the possibility that different species classes shift by different amounts or in different directions between domains), nor can it correct non-linear differences in feature distribution. When the domain gap is large — as between the Krishna delta and Pichavaram, which differ considerably in latitude, estuarine hydrology, and canopy structure — a single global linear correction of this kind is insufficient to recover reliable class boundaries, even though it is highly effective at reducing the aggregate covariance gap, as shown quantitatively in Chapter 2. The same-domain Pichavaram baseline (Section 4.2.1) confirms that this is a domain-adaptation limitation rather than a feature-quality limitation: the same 64-band features that yield only 48.9% accuracy under cross-domain transfer yield over 98% accuracy when no domain shift is involved.

This pattern is consistent with a broader limitation noted in the domain adaptation literature: shallow alignment methods that reduce a single, aggregate measure of domain discrepancy can still suffer substantial negative transfer when the target domain's class-conditional structure differs from the source, since matching the marginal feature distribution does not guarantee that individual classes are correctly aligned to their corresponding source classes. Li and Chen (2022)

similarly report that unsupervised domain adaptation approaches based on aligning entire domain distributions are prone to a "mode collapse" effect, in which alignment methods degrade performance because the target domain's multi-modal class structure is not respected by a single global transformation.

4.5 Chapter Summary

This chapter presented the cross-domain classification results for the Krishna → Malad and Krishna → Pichavaram transfer tasks. CORAL-based domain adaptation achieved a strong result on Malad (86.2% OA, $\kappa = 0.554$) but a substantially weaker result on Pichavaram (48.9% OA, $\kappa = 0.196$), despite a same-domain Pichavaram baseline confirming that the underlying features remain highly discriminative (~98–99% OA). This contrast demonstrates both the practical utility and the fundamental limitations of shallow, linear, second-order statistical domain alignment for cross-region ecological mapping, and motivates the exploration of more expressive adaptation strategies discussed in Chapter 5.

Chapter 5

Conclusion and Future Scope

5.1 Conclusions

This study set out to evaluate unsupervised domain adaptation for cross-region mangrove species classification, using pre-processed 64-band feature embeddings covering the Krishna delta (source domain) and the Malad and Pichavaram mangrove regions (unlabelled target domains). Four shallow domain alignment strategies — Subspace Alignment, Correlation Alignment, and their two sequential hybrids — were first benchmarked on a controlled Krishna 2019→2024 temporal domain-shift task, from which CORAL emerged as the most effective method, achieving the strongest joint reduction in covariance gap and subspace distance. CORAL was then applied to the substantially harder cross-region transfer task, in which a Krishna-trained one-vs-rest Random Forest classifier was used to classify mangrove species in Malad and Pichavaram. The Krishna→Malad transfer achieved 86.2% overall accuracy (κ = 0.554), while the Krishna→Pichavaram transfer achieved only 48.9% (κ = 0.196) — a same-domain Pichavaram baseline of ~98–99% confirmed that this gap in performance reflects the magnitude of the domain shift rather than any deficiency in the underlying feature representation.

5.2 Contributions of the Study

The principal contributions of this internship project are: (i) a quantitative comparison of four shallow domain alignment techniques — SA, CORAL, CORAL→SA, and SA→CORAL — on a real-world, high-dimensional mangrove feature dataset, using covariance gap and subspace distance as evaluation criteria; (ii) a stable-pixel selection procedure based on cosine-distance thresholding, used to robustly estimate CORAL alignment statistics under partial temporal land-cover change; (iii) a complete cross-domain mangrove species classification framework combining CORAL-based domain alignment with one-vs-rest Random Forest classification, probability map generation, and water masking; and (iv) an empirical demonstration — via a same-domain baseline comparison — that distinguishes domain-adaptation failure from feature-quality failure in cross-region ecological mapping, a distinction that is often left unexamined in applied domain adaptation studies.

5.3 Limitations

Several limitations of the present approach should be acknowledged. First, CORAL, SA, and their hybrids are shallow, linear alignment methods: they can only correct domain shift that is expressible as a linear transformation of the feature space, and have no mechanism for modelling class-conditional shift, in which different species classes are displaced differently between domains. Second, being fully unsupervised with respect to the target domain, these methods have no means of verifying — during alignment itself — whether the resulting transformation preserves class-discriminative structure; alignment quality was assessed only after the fact, via covariance gap, subspace distance, and (where available) target ground truth. Third, the classification accuracy achieved on Pichavaram (48.9%) demonstrates that CORAL's effectiveness is highly sensitive to the magnitude of the domain gap, and is not guaranteed to generalise to arbitrary target regions. Fourth, the accuracy assessment for the cross-domain experiments relied on the ground-truth samples available for each target region, which may not be fully representative of the entire study area. Finally, the 64-band feature embeddings used throughout this study were supplied as pre-processed input as part of the internship project; the sensitivity of the reported results to the specific feature representation used has not been independently evaluated within this work.

5.4 Future Scope

Building on the limitations identified above, several directions are proposed for future work: (i) evaluating deep, non-linear domain adaptation approaches — such as domain adversarial neural networks or class-conditional alignment methods — that may better accommodate the larger domain gap observed for Pichavaram; (ii) incorporating a pseudo-labelling or self-training stage, in which high-confidence target-domain predictions are iteratively used to refine the alignment, in the manner of progressive subspace refinement approaches described in the domain adaptation literature; (iii) extending the cross-domain evaluation to additional Indian mangrove regions beyond Malad and Pichavaram, to characterise more systematically how CORAL's effectiveness varies with the magnitude of inter-region ecological and sensor-driven domain shift; and (iv) incorporating raw NISAR L-band SAR observations directly into the feature representation as the mission's science-quality data products become available, to assess whether SAR-derived structural information improves cross-domain transferability relative to the pre-processed feature embeddings used in this study.

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Appendices

Appendix A — CORAL Implementation

The CORAL domain alignment procedure was implemented in Python, following the closed-form transformation of Section 2.2. The key steps of the implementation are summarised below:

1. Compute the source and target covariance matrices, C_S and C_T, from the stable-pixel-filtered feature samples (Section 2.5.1).
2. Add a ridge regularisation term ($\varepsilon = 1\times10^{-5}$) to each covariance matrix to ensure numerical stability during matrix inversion.
3. Compute C_S^(-1/2) and C_T^(1/2) via eigendecomposition of the respective (regularised) covariance matrices.
4. Apply the whitening–recolouring transform to the source (or target) features as given in Section 2.2, producing the CORAL-aligned feature set used for classifier training or inference.

Appendix B — Class Distribution in the Krishna Ground-Truth Dataset

Table B.1 lists the class-wise pixel counts in the Krishna delta ground-truth shapefile used as the labelled source domain throughout this study.

Class ID	Label	Pixels
1	A_marina_dense	2115
2	A_marina_moderate	856
3	A_marina_sparse_rej	2003
4	E_aggalocha	110
5	B_gymnorhiza	85
6	Rhizophora_sp	545
7	RH_BG_AM	443
8	MIXED	81
9	Sparse_mangrove	879
10	Prosopis	2098
11	SB_Suaeda_sp	1277
12	Creek_Waterbody	1559
13	Mudflat	2643
14	Sand_bars	2004

Table B.1: Class-wise pixel counts in the Krishna delta ground-truth dataset.